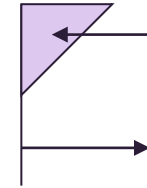
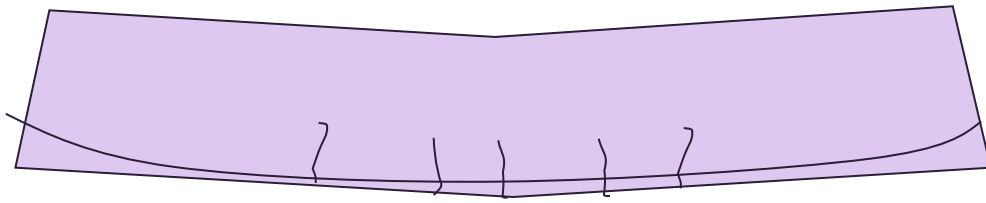
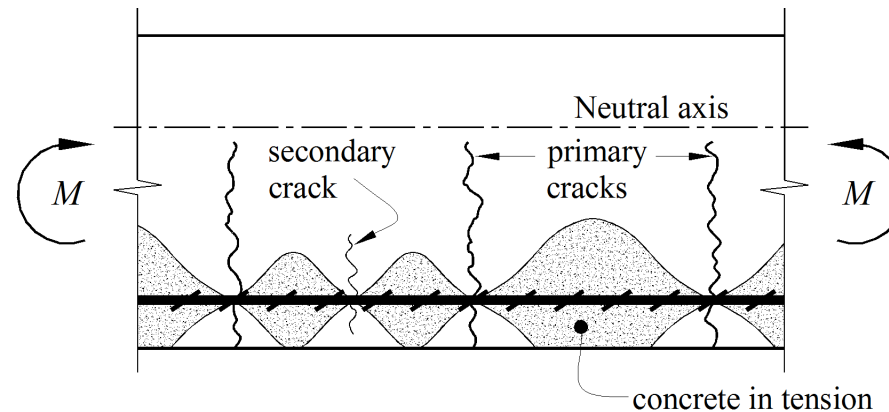


Design of Prestressed Concrete (PC) Beams for Serviceability



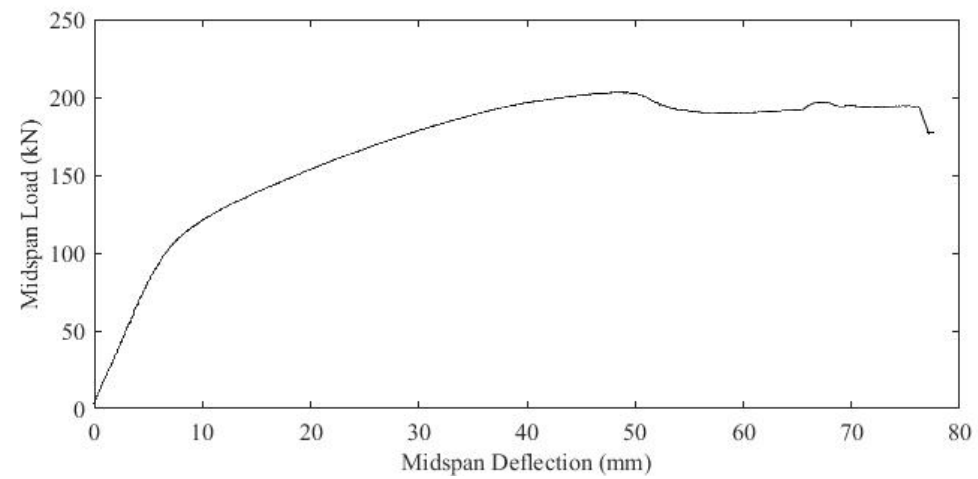
$$I \neq I_g, \quad A \neq A_g$$





North Carolina State University
Constructed Facilities Laboratory





Cracking moment, Clause 8.5.3.1 of AS3600

Applied moment at which the bottom fibre of the beam reaches the tensile cracking stress of concrete

$$M_{cr} = Z(f'_{ct.f} - \sigma_{cs} + P / A_g) + Pe \geq 0$$

Z = section modulus uncracked section I_g / y

σ_{cs} = maximum shrinkage induced tensile stress on uncracked section at the extreme fibre at which cracking occurs taken as

$$\sigma_{cs} = E_s \varepsilon_{cs}^* (2.5 p_w - 0.8 p_{cw}) / (1 + 50 p_w)$$

Web reinforcement ratio for tensile reinforcement

$$p_w = (A_{st} + A_p) / (b_w d)$$

Web reinforcement ratio for compressive reinforcement

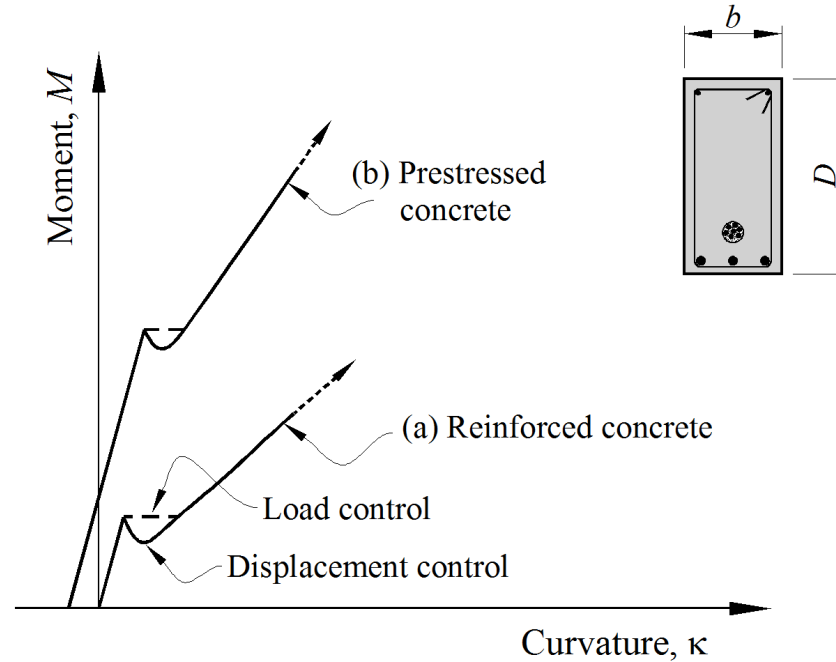
$$p_{cw} = A_{sc} / b_w d$$

ε_{cs}^* final design shrinkage strain of concrete

$f'_{cf.f}$ = flexural tensile strength of concrete

$$f'_{ct.f} = 0.6 \sqrt{f'_c}$$

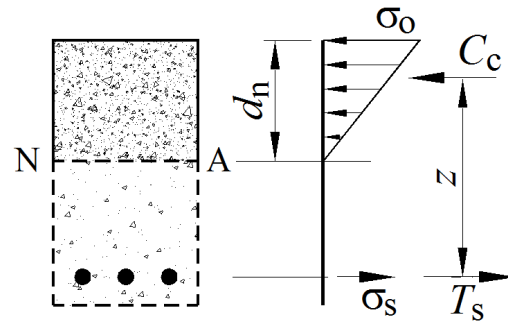
Transition from uncracked to cracked conditions



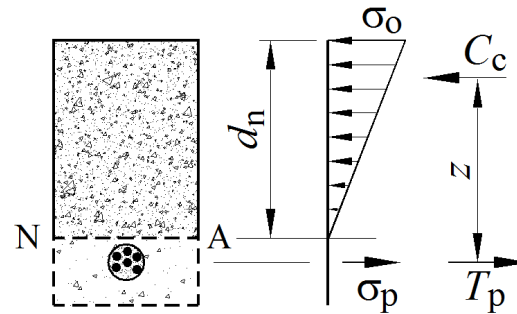
- Sudden loss in concrete area in tensile zone of RC or PC section at first cracking
- Section loses flexural stiffness during changeover from uncracked to cracked section
- Presence of eccentric prestress in section provides for an initial negative curvature

Post cracking behaviour

Post cracking stress distribution of a section



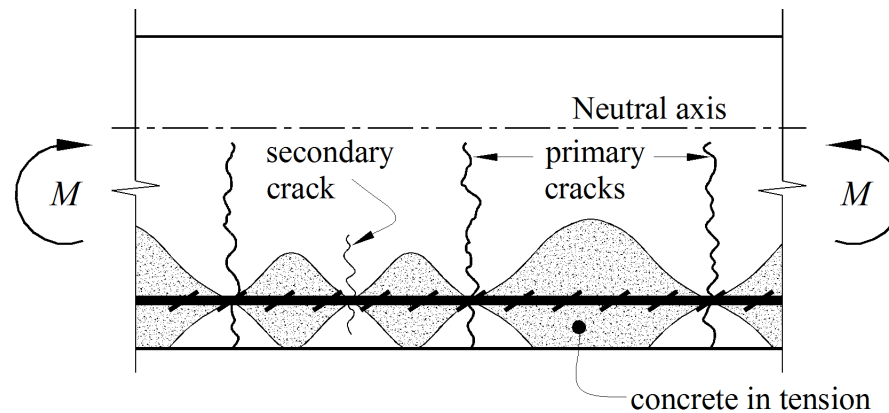
(a) Reinforced concrete
(d_n, z constant)



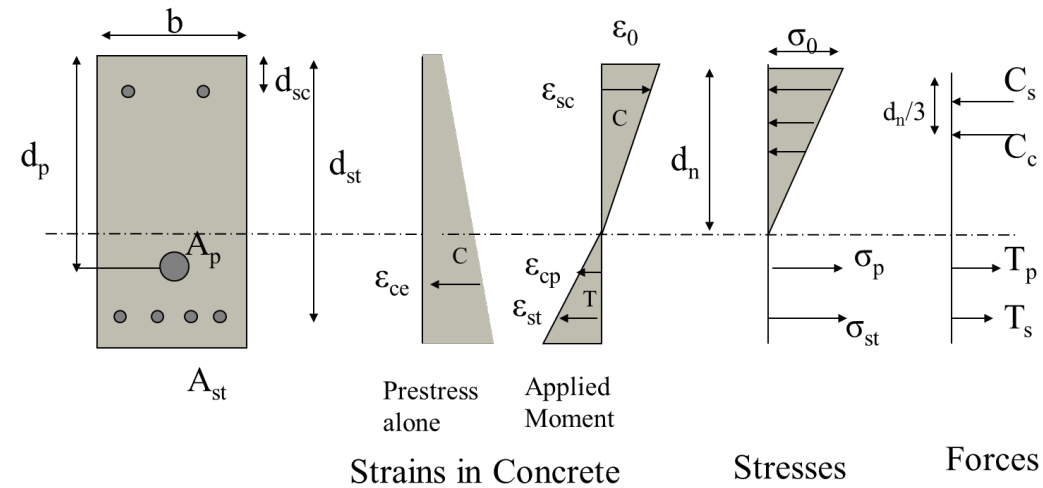
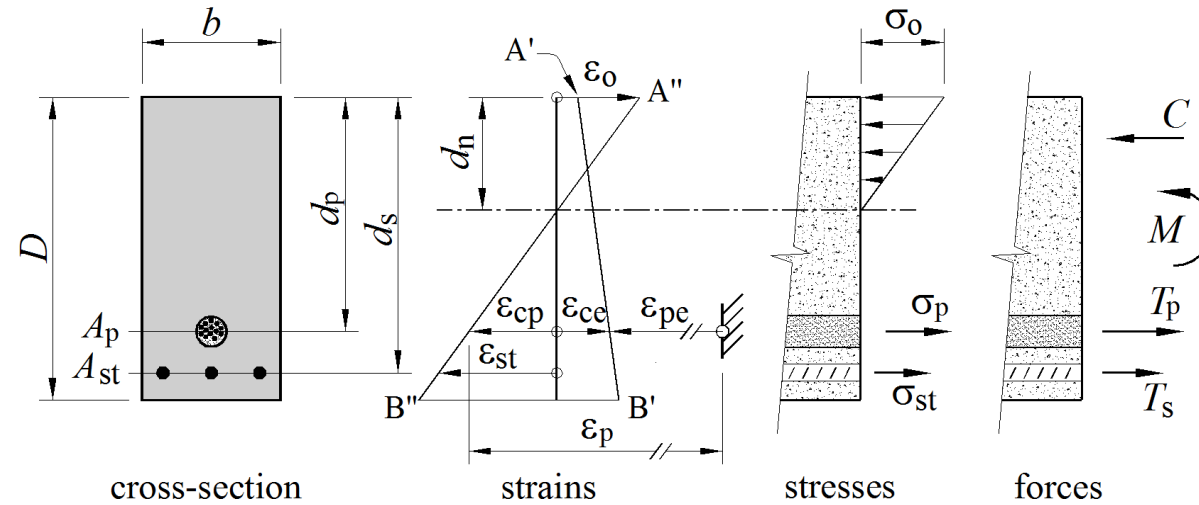
(b) Prestressed concrete
(d_n, z vary with M)

- Level of prestress effects post cracking behaviour
- N/A position and crack tip rise with increase in applied moment

- Flexural cracks and tension stiffening regions



Elastic Analysis of a rectangular cracked section



Equations of equilibrium for the section

Force Equilibrium

$$C_c + C_s = T_p + T_s$$

Moment Equilibrium,

Taking moments about the top of the section, (anti-clockwise is +ve)

$$M = T_s d_{st} + T_p d_p - C_c d_n/3 - C_s d_{sc}$$

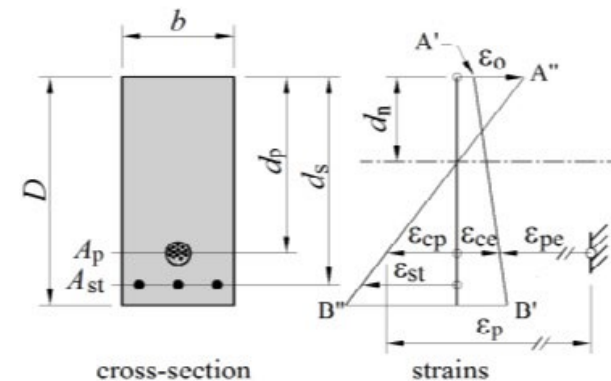
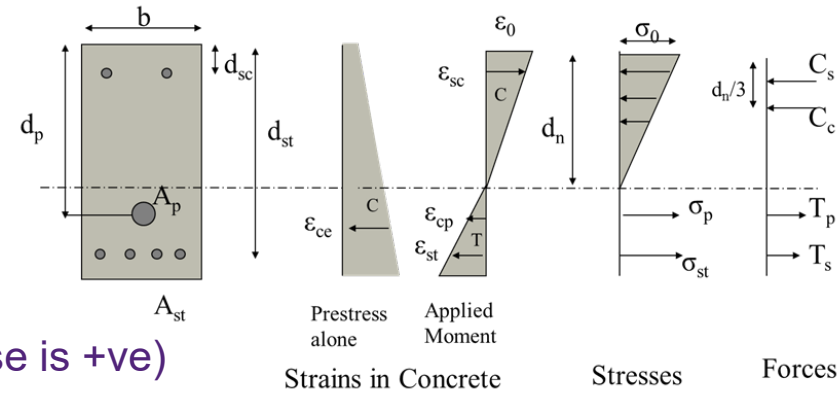
(line of action C_c is $d_c = d_n/3$)

Where 'M' is the applied external moment

Tensile strain in the concrete at the levels of the reinforcement and prestressing steel

$$\epsilon_{st} = \epsilon_{cst} = \epsilon_0 (d_{st} - d_n) / d_n$$

$$\epsilon_{cp} = \epsilon_0 (d_p - d_n) / d_n$$



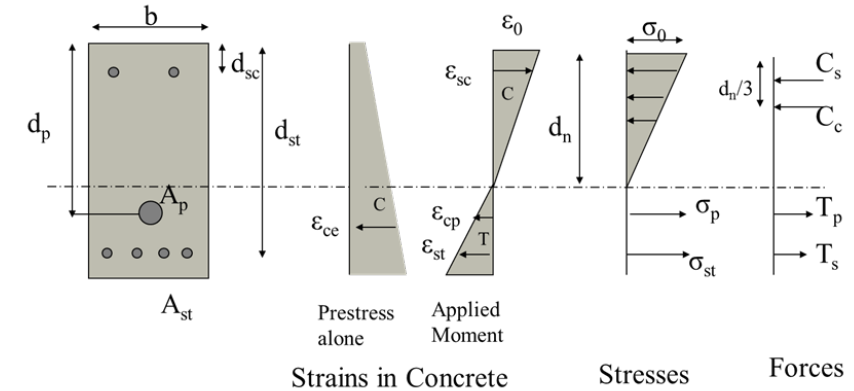
Calculation of forces in the section

$$C_c = 0.5\sigma_0 b d_n = 0.5E_c \epsilon_0 b d_n$$

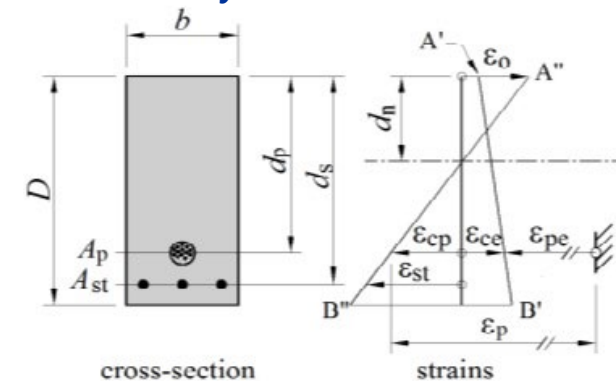
$$C_s = A_{sc} E_s \epsilon_{sc} = A_{sc} E_s \epsilon_0 (d_n - d_{sc}) / d_n$$

$$T_s = A_{st} \sigma_{st} = A_{st} E_s \epsilon_{st} = A_{sc} E_s \epsilon_0 (d_{st} - d_n) / d_n$$

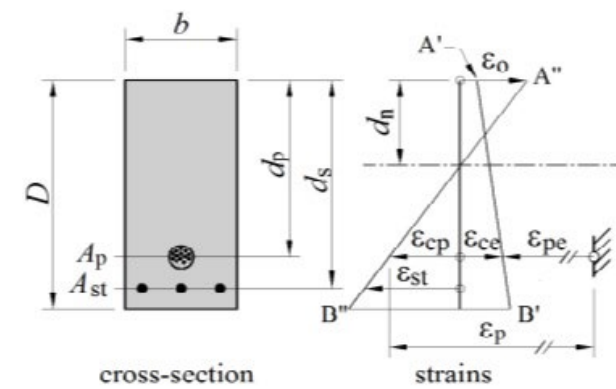
$$T_p = A_p \sigma_p = A_p E_p \epsilon_p$$



For a given strain diagram of concrete, all the parameters can be established. Only tricky issue is establishing ϵ_p .



Strain in prestressing steel



due to effective prestress

$$\varepsilon_{pe} = \sigma_p / E_p = P / (A_p E_p)$$

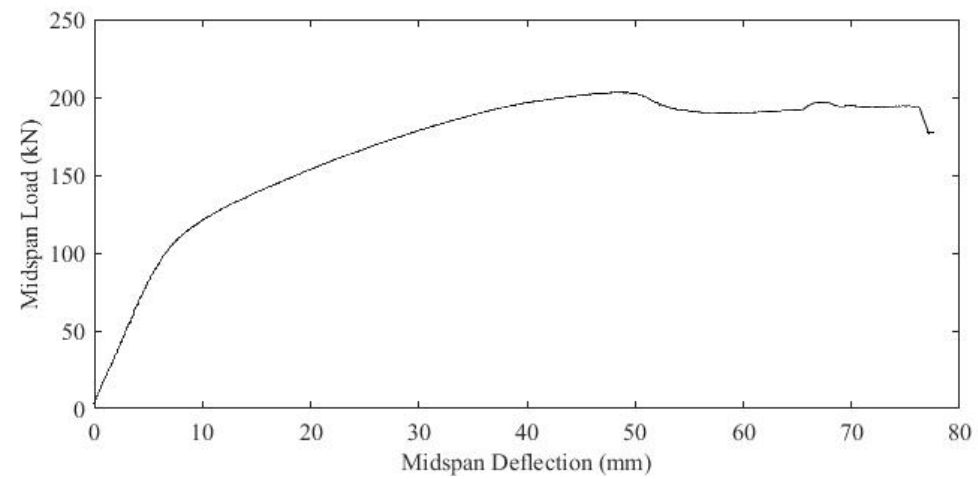
Compressive stress in concrete adjacent to the prestressing cable, due to prestress alone (use $P = P_e$ = effective prestress force)

$$\varepsilon_{ce} = \sigma_{ce} / E_c \text{ where } \sigma_{ce} = P / A_g + P e^2 / I_g$$

Tensile strain in concrete at level of prestress steel due to applied Moment

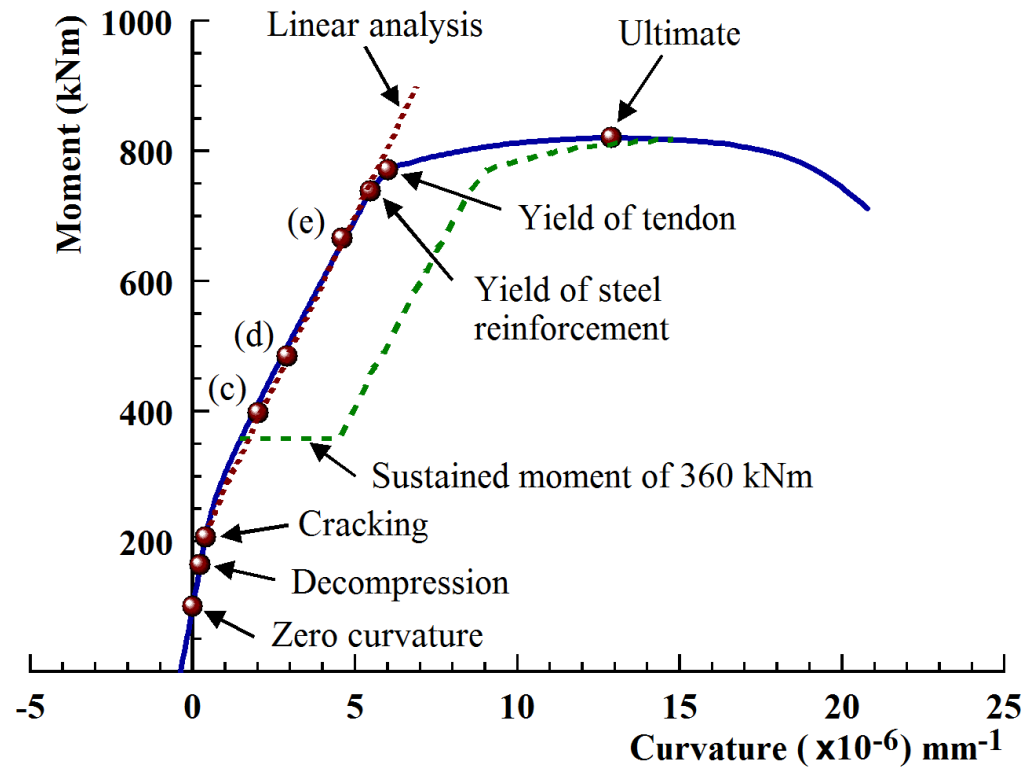
$$\varepsilon_{cp} = \varepsilon_0 (d_p - d_n) / d_n$$

$$\text{Total strain in prestress steel} = \varepsilon_p = \varepsilon_{pe} + \varepsilon_{ce} + \varepsilon_0 (d_p - d_n) / d_n$$

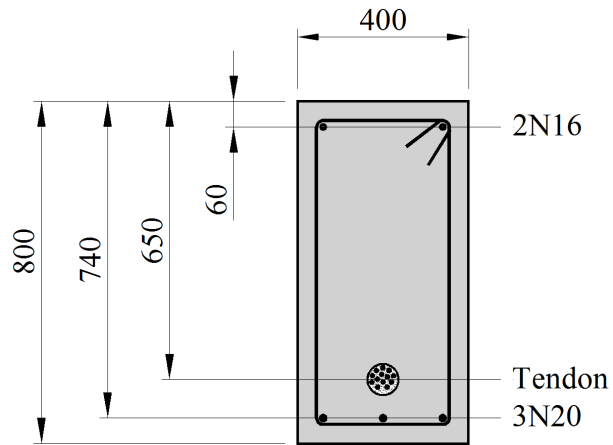


Moment-Curvature relationship

- Moment curvature relationships provide a good indication of the overload behaviour of flexural members
- Can also be used to calculate deflections under overload conditions



Moment-Curvature relationship – effect of p_p



$$d_{sc} = 60 \text{ mm}$$

$$d_p = 650 \text{ mm}$$

$$d_{st} = 740 \text{ mm}$$

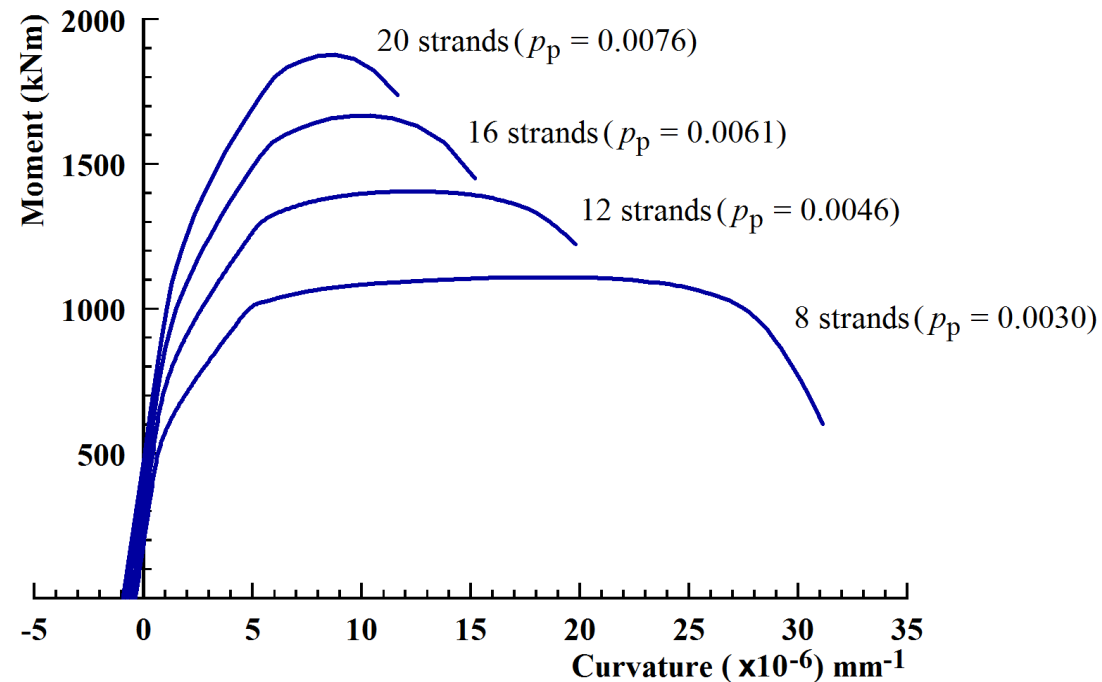
$$p_s = A_{st} / (bd_s) = 0.0031$$

$$p_p = A_p / (bd_p)$$

$$f'_c = 40 \text{ MPa}$$

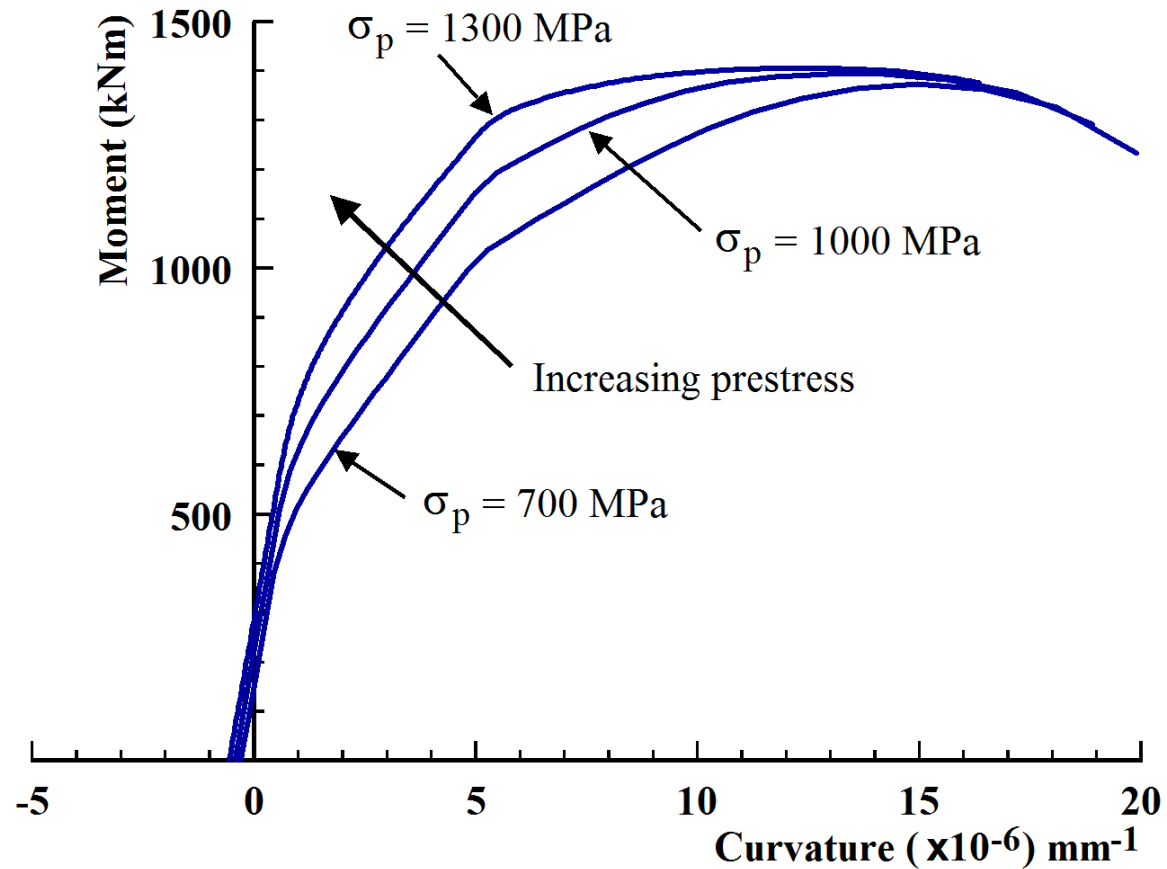
$$12.7 \text{ mm strand}$$

$$\sigma_p = 1300 \text{ MPa}$$



Moment-Curvature relationship – effect of stress σ_p

Fixed steel quantities



Short –term deflections uncracked member

Fully PC members, gross section properties are used

Deflection curve for flexural member at any stage of loading obtained from double integration of the curvature taking into account boundary conditions

$$y(x) = \iint \kappa(x) dx dx$$

Curvature $\kappa(x)$ at cross section along the beam depends on the bending moment in that section $M(x)$

Uncracked elastic deformations based on E_c and I_g

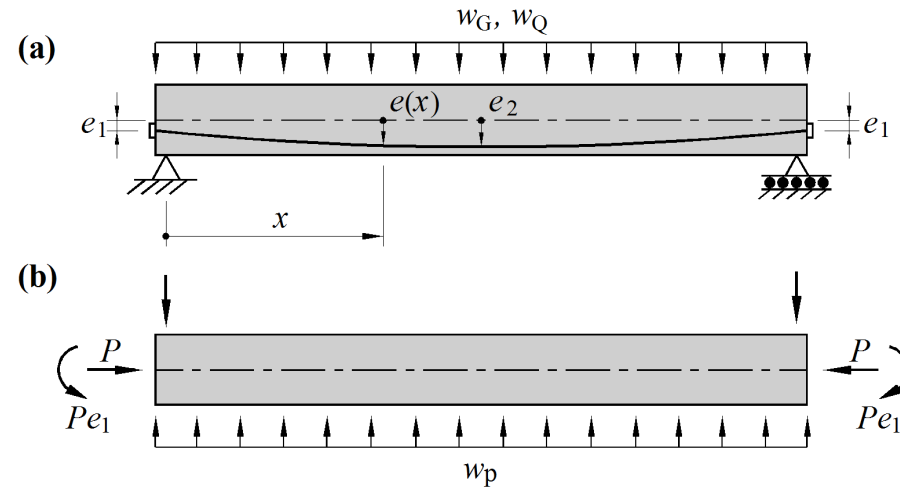
$$\kappa(x) = \frac{M(x)}{E_c I_g(x)}$$

Hence elastic deformations for uncracked members can be calculated using this general approach

$$y(x) = \iint \frac{M(x)}{E_c I_g(x)} dx dx$$

Initial Deflections

Elastic deformation calculated by standard formulae and coefficients



- Due to prestress equivalent upward load w_p

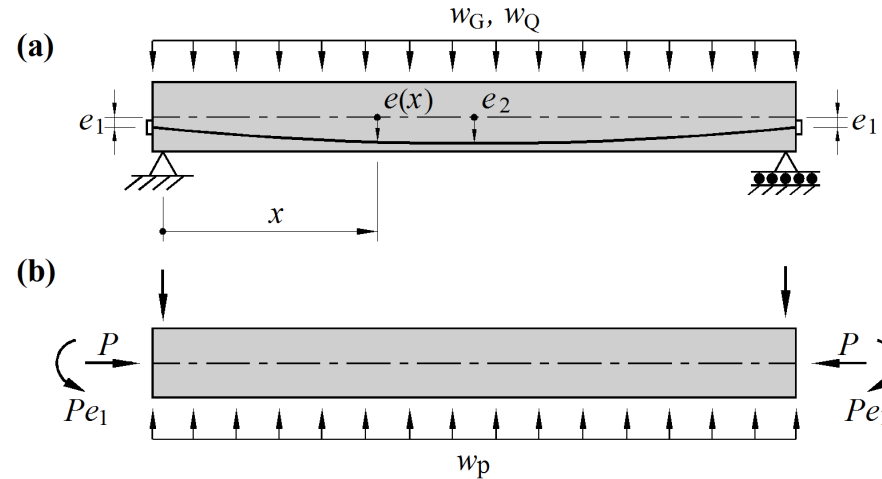
$$w_p = \frac{8Ph}{L^2} \text{ where } h = \text{sag} = (e_2 - e_1)$$

$$\Delta_p = \frac{5}{384} \frac{w_p L^4}{E_c I_g} (\text{upwards}) \uparrow$$

- Due to constant end moment Pe_1

$$\Delta_1 = \frac{1}{8} \frac{Pe_1 L^2}{E_c I_g} (\text{upwards}) \uparrow$$

Initial Deflections



- Due to dead load w_G

$$\Delta_G = \frac{5}{384} \frac{w_G L^4}{E_c I_g} (\text{down}) \downarrow$$

- Due to live load w_Q

$$\Delta_Q = \frac{5}{384} \frac{w_Q L^4}{E_c I_g} (\text{down}) \downarrow$$

- Total short term deflection

$$\Delta_{GQp} = \Delta_G + \Delta_Q - \Delta_p$$

$$\Delta_{initial} = \Delta_G + \psi_s \Delta_Q - \Delta_p$$

Short –term deflections cracked member Cl8.5.3.1

Partially prestressed concrete members

Deflection curve for flexural member at any stage of loading obtained from double integration of the curvature taking into account boundary conditions

$$y(x) = \iint \kappa(x) dx dx$$

Curvature $\kappa(x)$ for cracked section depends on the bending moment in that section $M(x)$ and flexural stiffness EI

Modifications can be made for tension stiffening to predict more accurate average curvature.

$$\kappa(x) = f[M(x)]$$

A simplified method for calculating deflections in cracked prestressed members uses the 'effective flexural stiffness $E_c I_{ef}$ (given in AS 3600 Cl. 8.5.3)

$$I_{ef} = I_{cr} + (I_g - I_{cr}) \left(\frac{M_{cr}}{M_s^*} \right)^3 \leq I_{ef, \max}$$

The effective section moment of area I_{ef} is defined in AS3600 Cl 8.5.3.1 (c)

This is known as Branson's equation and requires cracked section analysis to calculate I_{cr}

M_s^* the maximum bending moment at the section based on short-term serviceability load

$$M_{cr} = Z(f'_{ct.f} - \sigma_{cs} + P / A_g) + Pe \geq 0$$

$I_{ef, \max}$ = maximum effective second moment of area and is taken as I_g when $A_{st}/db \geq 0.005$

or $0.6I_g$ when $A_{st}/bd < 0.005$

NOTE: If member is fully prestressed then $I_{ef} = I_g$

Long-term deflections Cl 8.5.3.2

Sum of

- Shrinkage component of long-term deflections based on design shrinkage strain of concrete ε_{cs} (AS3600 Cl3.1.7) and principle of mechanics
- additional long-term creep deflections, determined from the design creep coefficient ϕ_{cc} (AS3600 Cl3.1.8) and principle of mechanics

In the absence of more accurate calculations additional long term deflection due to concrete creep and shrinkage may be calculated by multiplying short term deflection caused by sustained loads by a multiplier, k_{cs}

Long-term deflections

Additional long term deflection

$$\Delta_{LT} = \Delta_{pg} k_{cs}$$

k_{cs} (Cl. 8.5.3.2) and Δ_{pg} deflection due to applied prestress and self weight only (ie sustained loads)

$$k_{cs} = [2 - 1.2(A_{sc} / A_{st})] \geq 0.8$$

- Total long term deflection

$$\Delta_{total} = \Delta_{initial} + \Delta_{LT}$$

under short term service loads

$$\Delta_{initial} = \Delta_G + \psi_s \Delta_Q - \Delta_p$$

Deflections are then checked against suggested limits given in AS 3600 T2.3.2

TABLE 2.3.2
LIMITS FOR CALCULATED VERTICAL DEFLECTIONS
OF BEAMS AND SLABS

Type of member	Deflection to be considered	Deflection limitation (Δ/L_{eff}) for spans (Notes 1 and 2)	Deflection limitation (Δ/L_{eff}) for cantilevers (Note 4)
All members	The total deflection	1/250	1/125
Members supporting masonry partitions	The deflection that occurs after the addition or attachment of the partitions	1/500 where provision is made to minimize the effect of movement, otherwise 1/1000	1/250 where provision is made to minimize the effect of movement, otherwise 1/500
Members supporting other brittle finishes	The deflection that occurs after the addition or attachment of the finish	Manufacturer's specification but not more than 1/500	Manufacturer's specification but not more than 1/250
Members subjected to vehicular or pedestrian traffic	The imposed action (live load and dynamic impact) deflection	1/800	1/400
Transfer members	Total deflection	1/500 where provision is made to minimize the effect of deflection of the transfer member on the supported structure, otherwise 1/1000	1/250

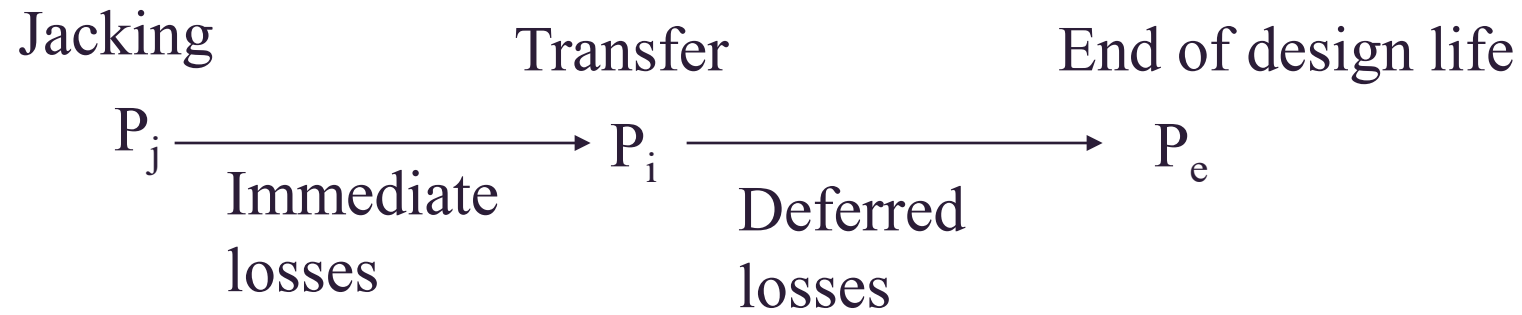
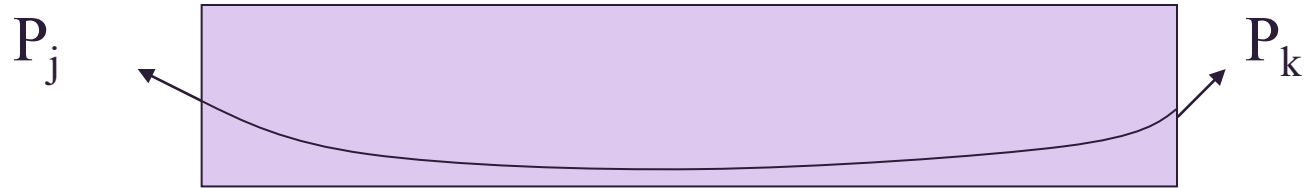
Two critical stages in the design of PC beams for serviceability

1st stage is immediately after the prestressing force P_i is transferred to the concrete.

2nd stage is after time-dependent losses have occurred and the member is under full service loads. At this stage, the magnitude of the prestressing force P_e , is at a minimum value and the applied loads at a maximum value.

At both these stages serviceability requirements of the members must be satisfied

Losses of Prestress



- Elastic deformation of concrete
- Anchorage Slip
- Duct Friction

- Stress relaxation of prestressing steel
- Shrinkage deformation of concrete
- Creep deformation of concrete

Serviceability limit state design of PC members – crack control

Design at transfer P_i .

Only immediate losses have occurred. Member is subject to the minimum applied loading ie self-weight and prestress only.

Minimum allowable concrete strength at transfer = f_{cp}

The determines the upper limit of the magnitude of the prestressing force

AS3600 Clause 8.1.6.2 provides a ‘deemed to satisfy’ criterion if the maximum compressive stress in the concrete at transfer is limited to $0.5f_{cp}$

Serviceability limit state design of PC members – crack control (cont'd)

Design for stress limit under full service load

Fully prestressed members are required to remain uncracked under full service load, and so maximum tensile stresses $\leq 0.25 \sqrt{f'_c}$ Clause 8.6.3

For partially PC members, some cracking is permitted and Cl 8.6.3 requires:

reinforcement to be provided near the tensile face

the calculated maximum flexural tensile stress under short term service loads to $\leq 0.6 \sqrt{f'_c}$ to control cracking in a 'deemed to satisfy' provision

Worked example- Deflection calculation for simply supported beam

Calculate the short-term and long-term deflection for the simply supported post-tensioned beam which spans 10m and has a parabolic tendon with an eccentricity at midspan of 250mm and zero at the ends.

The initial cable force, P_i is 1200kN and the live load is $w_q = 30\text{kN/m}$ $w_g = 8\text{kN/m}$. Take deferred losses = 10%

(a) Check the adequacy of the beam at transfer using the provisions of Cl. 8.1.6.2

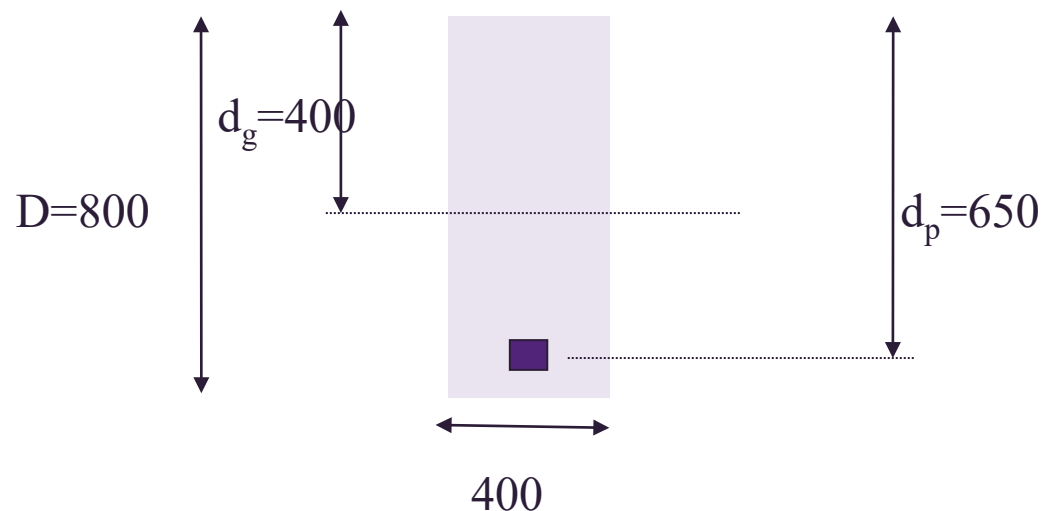
(b) Calculate the deflection of the beam.

-short term deflection at transfer

-short term deflection under service loads

-additional long term deflection

-total long term deflection



$$\begin{aligned}f'_{cp} &= 32 \text{ MPa} \\f'_c &= 40 \text{ MPa} \\P_e &= 1080 \text{ kN} \\P_i &= 1200 \text{ kN} \\A_p &= 1000 \text{ mm}^2 \\A_g &= 320,000 \text{ mm}^2 \\I_g &= 17.07 \times 10^9 \text{ mm}^4 \\E_c &= 32800 \text{ MPa} \\E_p &= 200,000 \text{ MPa} \\Z_{bot} &= I_g / y_{bot} \\&= 17.07 \times 10^9 / 400 \\&= 42.67 \times 10^6 \text{ mm}^3\end{aligned}$$

Worked example- Deflection calculation for simply supported beam

Calculate the short-term and long-term deflection for the simply supported post-tensioned beam which spans 10m and has a parabolic tendon with an eccentricity at midspan of 250mm and zero at the ends.

The initial cable force, P_i is 1200kN and the live load is $w_q = 30\text{kN/m}$ $w_g = 8\text{kN/m}$. Take deferred losses = 10%

(a) Check the adequacy of the beam at transfer using the provisions of Cl. 8.1.6.2

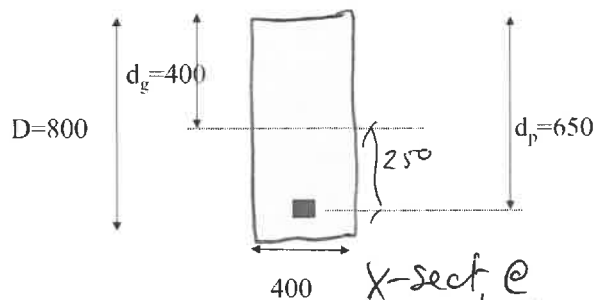
(b) Calculate the deflection of the beam.

-short term deflection at transfer

-short term deflection under service loads

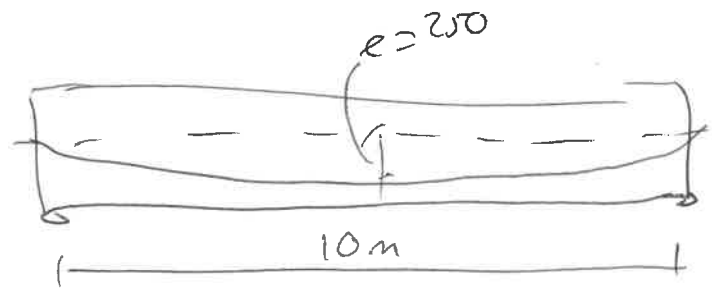
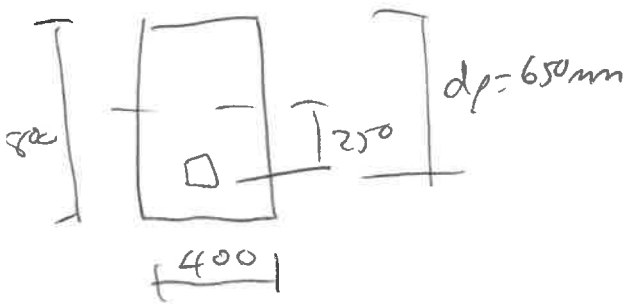
-additional long term deflection

-total long term deflection



$$\begin{aligned} f'_{cp} &= 32 \text{ MPa} \\ f'_c &= 40 \text{ MPa} \\ P_c &= 1080 \text{ kN} \\ P_i &= 1200 \text{ kN} \\ A_p &= 1000 \text{ mm}^2 \\ A_g &= 320,000 \text{ mm}^2 \\ I_g &= 17.07 \times 10^9 \text{ mm}^4 \\ E_c &= 32800 \text{ MPa} \\ E_p &= 200,000 \text{ MPa} \\ Z_{bot} &= I_g / y_{bot} \\ &= 17.07 \times 10^9 / 400 \\ &= 42.67 \times 10^6 \text{ mm}^3 \end{aligned}$$

Lecture Example



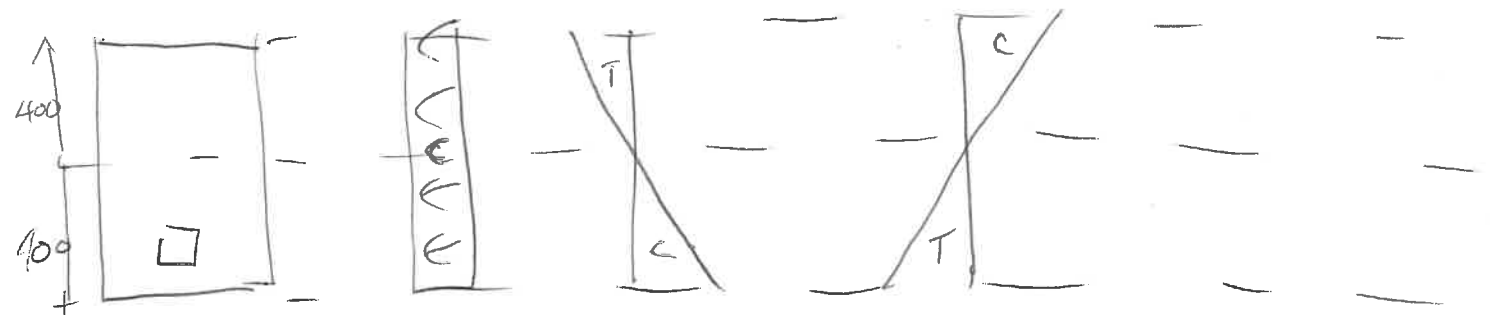
a) Check the adequacy of the beam at transfer using provisions of CI 8.1.6.2.

$$f_{cp} = 32 \text{ MPa} \quad P_i = 1200 \text{ kN} \quad A_p = 1000 \text{ mm}^2$$

$$A_g = 320,000 \text{ mm}^2 \quad I_g = 17.07 \times 10^9 \text{ mm}^4$$

M_g = self-weight moment span = 10 m

$$w_g = 0.32 \times 25 \text{ kN/m} = 8 \text{ kN/m} \quad M_g = \frac{w_l^2}{8} = 100 \text{ kNm}$$



$$\sigma_b = \frac{P_i}{A_g} + \frac{P_i \cdot e \cdot y_b}{I_g} - \frac{M_g \cdot y_b}{I_g} \quad (N_{mm} \text{ MPa})$$

$$\sigma_b = \frac{1200 \times 10^3}{320,000} + \frac{1200 \times 10^3 \times 250 \times 400}{17.07 \times 10^9} - \frac{100 \times 10^6 \times 400}{17.07 \times 10^9}$$

$$= 3.75 + 7.03 - 2.34$$

$$= 8.44 \text{ MPa} \leq 0.6 f_{cp} (0.6 \times 32) = 17.5 \text{ MPa}$$

$\therefore$ OK

b) i) short-term deflection @ transfer.

↓ ↓ ↓ ↓ ↓ ↓ ↓ $\omega g = 8 \text{ km/h} \downarrow$



$\uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \quad \omega_p = \frac{8\pi\epsilon_0 h}{L^2} =$

$$w_p = \frac{8 \times 1200 \times 0.25}{10^2} = 24 \text{ kN/m} \uparrow$$

$$\text{net load on beam} = 24 \text{ kN/m } \uparrow - 8 \text{ kN/m } \downarrow = 16 \text{ kN/m } \uparrow$$

$$\Delta_{ST} @ \text{transfer} = \Delta_{pg} = \frac{5 \omega L^4}{384 E_c \times I_g}$$

Take $\omega = \omega_{\text{net}} = 16 \text{ kW/m} \uparrow$.

$$\Delta p_{\text{e transfer}} = \frac{5 \times (-16) \times (10,000)^4}{384 \times 30,100 \times 17.07 \times 10^9} = -4.1 \text{ mm} \quad \uparrow$$

(upward camber)

(ii) short-term deflection due to service loads

$$P_e = 1080 \text{ kN}$$

$$\psi_{ST} = 0.7$$

$$f_{tk} = 40 \text{ MPa}$$

$$E_c = 32,800 \text{ MPa}$$

$$W_{service} = W_g + \psi_{ST} W_q$$

$$= 8 + 0.7 \times 30 = 29 \text{ kN/m}$$

$$M_s = \frac{W l^2}{8} = \frac{29 \times 10^2}{8} = 362 \text{ kNm.}$$

@ mid-span

1st calculate $M_{cr} = Z \left(f_{ctf} - \sigma_{cs} + \frac{P_e}{A_g} \right) + P_e \cdot e$

→ Cl 8.5.3.1

σ_{cs} = maximum shrinkage induce prestress

$$= E_s \cdot \epsilon_{cs}^* \left(\frac{2.5 p_w - 0.8 p_w}{1 + 50 p_w} \right)$$

$$p_w = 0$$

$$\epsilon_{cs} = 565 \times 10^{-6} - \text{cl 3.1.7}$$

= design shrinkage strain.

$$p_w = \frac{A_{pt}}{b d} = \frac{1000}{600 \cdot (400 \times \frac{650}{650})} = 0.0038$$

$$\sigma_{cs} = 200 \times 10^3 \times 565 \times 10^{-6} \left(\frac{2.5 \times 0.0038}{1 + 50 \times 0.0038} \right)$$

$$= 0.91 \text{ MPa.}$$

$$f_{ctf} = 0.6 \sqrt{f_{tk}} = 4 \text{ MPa}$$

$$M_{cr} = Z \left(f_{ctf} - \sigma_{cs} + \frac{P_e}{A_g} \right) + P_e \cdot e$$

$$= 42.7 \times 10^6 \left(4 - 0.91 + \frac{1080 \times 10^3}{320,000} \right) + 1080 \times 10^3 \times 25$$

$$= 534.5 \times 10^6 \text{ Nmm}$$

$$= 535 \text{ kNm} > M_s (362)$$

∴ uncracked section.

$$\Delta_{ST} = \frac{\sum}{384} \left(\frac{w_{net} \cdot L^4}{E_c \cdot I_g} \right)$$

$$w_p = \frac{8 \cdot l_e \cdot h}{L^2} = \frac{8 \times 1080 \times 0.25}{10^2} = 21.6 \text{ kN/m}$$

$$w_{net} = w_{sev} - w_p \\ = (29 - 21.6) = 7.4 \text{ kN/m} \downarrow$$

$$\Delta_{ST} = \frac{\sum}{384} \times \frac{7.4 \times 10,000^4}{32,800 \times 17.07 \times 10^9} \\ = 1.72 \text{ mm} \downarrow$$

(iii) additional long-term deflection.

$$\Delta_{LT} = \Delta_{pg} \cdot K_{cs} \quad \text{cl 8.5.3.2.}$$

$$K_{cs} = \left(2 - 12 \frac{A_{sc}}{A_{st}} \right) = 2.$$

$$w_{pg} = 8 \downarrow - 21.6 \uparrow = 13.6 \text{ kN/m} \uparrow$$

$$\Delta_{pg} = \frac{\sum}{384} \frac{-13.6 \times 10,000^4}{32,800 \times 17.07 \times 10^9} \\ = 3.16 \text{ mm} \uparrow$$

(iv) total long term deflection-

$$\Delta_{total} = (\Delta_{pg} + \phi_s \Delta_Q) - \Delta_p \\ = 1.72 \text{ mm} \downarrow$$

$$\Delta_{LT} = \Delta_{pg} \cdot K_{cs} \\ = 3.16 \times 2 = 6.32 \text{ mm} \uparrow$$

~~total~~

total deflection long term

$$= \Delta_{initial} + \Delta_{LT}$$

$$= 1.72\text{mm} \downarrow - 6.32\text{mm} \uparrow$$

$$= -4.6\text{mm} \uparrow.$$